

Smart Suction Cup: Enabling Tactile Feedback for Advanced Robotic Manipulation in Research and Recycling Applications



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EXECUTIVE SUMMARY

Born from the need to make robotic manipulation smarter, more responsive, and more sustainable, this Capstone project transforms a low-fidelity research prototype into a functional product that can be easily deployed for further development by robotics and AI communities. Our goal is to develop a fully open-source, tactile-enabled gripping system that combines vacuum-based adhesion with integrated pressure sensing and haptic feedback. The result is a system is capable of detecting contact conditions, grasp success, and slip events while autonomously correcting for them in real time. Designed for both academic and industrial research, the platform empowers exploration of vacuum suction cup technology for advanced robotic sorting tasks. Specifically, it supports research into automated trash-sorting, enabling robots to detect, grasp, and separate recyclables with precision, speed, and adaptability. The team has created a modular, plug-and-play gripper featuring universal robot compatibility, embedded control electronics, and real-time tactile sensing, turning complex lab technology into a user-friendly toolkit that accelerates innovation in robotic manipulation.

Over the course of the project, we benchmarked current robotic arm attachments and refined the Smart Suction Cup concept from a lab-only prototype into a robust end-of-arm tool that others can easily fabricate and reproduce. Key learnings include the value of modular mechanical interfaces, the use of compliant mechanisms to reduce complexity, and the importance of balancing compactness, manufacturability, and maintainability when designing replicable research hardware. Future work should focus on validating the system in industrial environments and expanding the open-source ecosystem around the gripper. This includes releasing benchmark datasets, example haptic controllers, and hardware variants for different payloads, allowing the Smart Suction Cup platform to advance research on tactile-enabled, feedback-corrective manipulation in recycling and beyond.

I. INTRODUCTION

Robotic manipulation in unstructured environments, such as recycling facilities, has transformed waste processing and research applications, with AI-enabled robots now being deployed in multiple material recovery facilities around the world [1]. However, existing end-effectors used in industrial robotic systems, such as those from Schunk or Robotiq, are often too expensive for widespread recycling deployment and generally lack integrated tactile sensing, limiting their performance and leaving average recycling contamination rates around 25% [2].

These standard vision-based grippers, whether suction cup based or not, struggle with adversarial objects, including: transparent or highly reflective items that confuse depth sensing and visual grasp planning, deformable or loosely packaged items that collapse under suction and break the seal, and porous or rough materials where air leaks through the surface, making it difficult to maintain vacuum. [3]

Overall, tactile-enabled robotic grippers, such as the Smart Suction Cup System the team at UC Berkeley’s Embodied

Dexterity Group (EDG) is working on, enable significantly higher grasping success rates on these adversarial objects, improving robustness in bin-picking and delivering more reliable haptic exploration. [4], [5], [6].

The aim of this project is to transform the research-grade Smart Suction Cup prototype from the EDG lab into a low-cost, compact, and modular end-of-arm tool, compatible with a variety of industrial robotic arms making it widely deployable. The improved design still uses multi-chamber flow sensing to provide haptic feedback at a lower cost than many other current gripper systems, and should provide tactile feedback that approaches the 2.5× grasp success improvements reported for the research-grade Smart Suction Cup in bin-picking experiments [5], [6].

II. BACKGROUND: RECYCLING AUTOMATION LANDSCAPE AND TACTILE ROBOTICS

A. How Recycling Facilities Automate Sorting Today

Modern material recovery facilities (MRFs) increasingly rely on layered automation pipelines rather than a single technology. Bulk separation equipment such as screens, ballistic separators, and air-classification systems first divide the incoming stream by size, which reduces the burden on downstream sorting [7], [8]. Sensor-based systems then perform finer separations: optical near-infrared (NIR) sorters identify plastics by polymer type, cameras distinguish paper grades, and induced current separators and magnetic drums remove ferrous metals from the stream [7], [9].



Fig. 1: Sorting Automation Methods

AI-powered systems have added a new decision-making layer on top of these conventional unit operations. Several vendors (e.g. AMP, TOMRA, ZenRobotics, EverestLabs) deploy computer-vision models above conveyor belts to identify individual items in mixed streams and command robotic arms to extract targeted objects [10], [11]. Industrial robotic grippers used in these systems are typically off-the-shelf parallel-jaw grippers, simple suction cups, or large vacuum arrays. They are optimized for robustness and low maintenance but generally lack precise sensing; feedback is often limited to vacuum pressure or motor current, which inhibits their ability to interact with cluttered objects, deformable items, and objects outside the training distribution of the vision system [12], [13]. In high-throughput container lines where installing large robots would be impractical,

AI-guided air-ejector systems are emerging as an alternative to robotic arms [14]. Still, many plants depend on manual pickers for quality control on critical lines, especially where contamination, unusual packaging, or changing product mixes challenge fully automated systems [15]. This gap between high-level perception (AI vision) and low-level contact information motivates research into tactile-enabled end-effectors that can recover from visual uncertainty and improve reliability on adversarial waste objects.

B. Tactile-Enabled Grippers Beyond the Smart Suction Cup

Outside of UC Berkeley, many research groups are exploring ways to embed tactile sensing directly into robotic grippers. Recent reviews of tactile sensing highlight a trend towards soft, multi-modal skins and fingertip sensors that can measure normal force, shear, slip, and local geometry, enabling more robust grasping and in-hand manipulation [16].

For example, several open-source parallel grippers integrate soft “skin” fingertips with embedded Hall-effect or piezoresistive elements, allowing three-axis force estimation and safer grasping of fragile items [17]. Other work has focused on optical tactile sensors such as GelSight or TacTip, which use internal cameras and deformable membranes to reconstruct high-resolution contact geometry; these have been embedded in gripper palms or fingertips to support precise pose adjustment and slip detection [18].

Soft robotic hands and hybrid grippers are another line of research. Soft pneumatic fingers combined with tactile palms have demonstrated robust grasps on irregular and delicate objects, achieving strong performance on a standardized set of objects and protocols for evaluating robotic manipulation tasks (the YCB benchmarks) [19]. Acoustic and capacitive tactile skins have also been proposed as low-cost overlays for existing industrial fingers, providing contact and slip cues without major hardware changes [20]. These systems show that integrating tactile sensing can significantly improve manipulation of deformable, slippery, or cluttered objects compared to purely vision-based control.

Our work sits at the intersection of these trends and the needs of recycling automation. Instead of designing a new multi-fingered hand, we focus on turning a multi-chamber suction concept into an open, modular, and low-cost end-effector that can be dropped into existing AI-enabled recycling cells. By combining vacuum grasping, which is already common in industry, with chamber-level pressure sensing; the resulting tool aims to deliver a practical path for MRFs and research labs to experiment with tactile feedback in real-world environments, complementing the more general-purpose tactile grippers developed in other domains.

III. DESIGNING AN OPEN-SOURCE, DEPLOYABLE SMART SUCTION CUP

A. Design Requirements and Criteria

To establish clear guidelines before starting the design phase, requirements are determined based on the performance goal and the stakeholder expectations. The require-

ments are divided into four subsections. The specifications include all precise requirements that are given at the start of the project. The functional requirements describe what the design should do. The preconditions determine specific goals that the design must adhere to, this gives insight into the limitations of the design. Lastly the performance criteria contain the specific standards that evaluate the quality of the design. To have a measurable goal for the multi-chambered suction cup its performance will be compared to current industrial suction grippers used in recycling cells, as well as to a baseline single-chamber cup mounted on the same robot [21]. The gripping performance of both grippers is measured in standardized pick-and-place experiments described in chapter III-B.

Specifications	
Compactness	Smaller than bounding box $80 \times 80 \times 60$ mm, but strictly smaller than the previous prototype.
Weight	Lighter than 1 kg, with center of mass as close as possible to the flange, such that the resulting inertia about the wrist does not significantly degrade dynamic performance.
Mountable	Interface complies with most common ISO 9409-1 robot flange standards for common industrial and cobot arms to ensure universal mounting.
Suction Cup	Tool-less installation of the silicon molded suction cup while providing a robust connection under the maximum specified load.
Compatible	Must be compatible with the existing vacuum supply interface.
Functional requirements	
Vacuum performance	System reliably maintains at least 80% vacuum under nominal conditions ($\leq 20kPa$ absolute).
Gripping force	The suction cup can provide a holding force corresponding to 1.2 times the nominal payload of 0.5 kg.
Detection	Must provide pressure feedback per chamber sufficient to detect grasp success or failure (sampling rate above 144Hz, control rate above 120Hz, and total latency from sensor measurement to control output < 10 ms).
Thermal Safety	External surfaces remain below $41^\circ C$ during normal operation (per IEC 60601-1 safe touch limits).
Reproducibility	Full CAD, PCB, and firmware sources plus a build guide will be published openly, and at least one external lab can reproduce the system within 1 working day of hands-on effort using only this documentation.
Cleaning	Components must be cleanable and safe according to industrial standards.
Preconditions	
Operation time	Maximal operating time of at least 60 minutes without maintenance intervention.
Durability	Must sustain at least 5000 picks before replacement of any suction-cup system part.
Mounting	Must be mounted on/off in 30 seconds.
Environment	Intended for indoor use between 10 and $35^\circ C$, non-condensing humidity to avoid obstructing vacuum.
(Dis)assembly	Can be taken apart and reassembled within 10 minutes.
Performance criteria	
Sustainability	As many parts as possible made from sustainable materials, with minimal embedded carbon footprint.
Grasp robustness	Must maintain a stable grasp under partial contact and small positional errors.
Production	Parts are easily manufactured using methods available in a standard robotics lab or maintenance workplace.
Cost	Target unit cost $\leq \$150$ per system. Lowest possible while meeting all other performance and reliability requirements.

TABLE I: Requirements and criteria

B. Performance Evaluation and Test Procedures

During prototyping and evaluation, the following measurable tests are defined to assess the performance of

the final prototype.

1) *Test 1: Maximum Payload and Safety Margin:*

- **Objective:** Determine the maximum payload mass that can be lifted and transported without loss of seal under nominal vacuum level.
- **Setup:** Suction cup mounted on a robot or linear axis, with calibrated test plates representing typical items in the recycling stream (e.g. plastic trays, bottles, metal lids, cardboard pieces) [8], [21].
- **Procedure:** Increase payload from 10 g to 1.5 kg and execute a standardized motion profile including vertical lifts and horizontal accelerations.

2) *Test 2: Setup Time and Interchangeability:*

- **Objective:** Evaluate the time and complexity required to assemble and configure the suction system from a disassembled state, as well as assess the ease of replacing mechanical and electronic components without specialized tools or recalibration [7], [21].
- **Setup:** All system components are provided separately. Operators follow standard assembly documentation and perform initial wiring, firmware upload, and calibration using a predefined workstation layout.
- **Procedure:** Record total setup time from unpacking to operational state, including mechanical assembly and software configuration. Repeat the procedure for three operators to assess variability. Additionally, record the time to replace one mechanical part from the assembly to evaluate interchangeability and maintenance accessibility.

3) *Test 3: Durability and Maintainability:*

- **Objective:** Assess wear and required maintenance over many picks in a dusty and abrasive environment [21], [22].
- **Setup:** Long-run endurance test with mixed items including abrasive objects and dust exposure, representative of material recovery facilities.
- **Procedure:** Run the system for $N = 6000$ picks while logging grasp success rate and pressure anomalies; inspect suction cup, tubing and fragile parts at regular intervals. Log the number of picks until first visible damage and time required for part replacement or cleaning.

C. Initial Design Choices: Chambers and Materials

Considering compactness and manufacturability, the cup body is implemented as a three-chamber design arranged symmetrically around the tool frame. This layout preserves directional redundancy for leak localization and partial-contact robustness, while keeping the internal fluidic network and external tubing simple enough for production in a typical university lab.

The suction cup is realized as a cast silicone cup using a reusable mold, providing compliant sealing on curved and slightly rough surfaces and sufficient durability for the

targeted pick counts. The vacuum tubing consists of off the shelf components. The casing and mounting hardware are 3D-printed from PLA, enabling rapid iteration, low-cost replication, and straightforward recycling of failed prints.

D. From Patchwork Electronics to Integrated Control PCB

The early prototypes of the Smart Suction Cup relied on separate "breakout boards" for the microcontroller, pressure sensors, and power regulators, which were connected by hand-wired jumpers and external cables. This "patchwork" architecture greatly increased the casing volume required, complicated the cable routing through the casing, and introduced many potential failure points during repeated mounting and cleaning.

To address this, we designed a custom two-layer control PCB that consolidates the Seeedstudio RP-2040 microcontroller, an I²C multiplexer, three digital pressure sensors, and power and communication interfaces onto a single board, which may be bulk ordered from PCB manufacturers. The integrated PCB simplifies assembly to a single board-to-cable connector, and exposes mounting holes and connectors near the perimeter for easy access during service. At current low order volumes, the cost is approximately \$80 per board when ordered preassembled, but with greater order volumes the projected costs lower to \$50 or less, keeping the system well within the \$150 target while providing high-quality sensing. The current design presents a scalability bottleneck. Specifically, the requirement for double-sided assembly significantly increases manufacturing costs, and the use of a discrete microcontroller breakout board requires manual soldering for each unit. To address this, the design must either be revised to integrate the breakout board functionality directly onboard of the PCB, or an alternative connection interface should be developed to eliminate the need for soldering during assembly.



Fig. 2: Cleaning Up The Electronics

E. Casing Concept Generation and Selection

After the material and early design choices were confirmed, the casing of our solution could be further developed around the geometry of the PCB.

The casing was first separated into two main areas of design: the suction cup attachment mechanism, and the general layout and geometry of the casing. For each of these areas, several potential solutions (10+) were generated and evaluated before selecting three for further development. For the suction cup attachment mechanism, the potential solutions developed were: twist-fit, where the suction cup twists into place, press-fit, where the suction cup is pressed directly in and held by friction, and compliant clip, where an external 3D printed clip is inserted and physically retains the suction cup. For the casing geometry, the three potential solutions were: fully-internal, where all tubing and electronics are internalized in a tube casing, box, where all components are contained within a rectangular box, and original which utilizes two halves which come together into a tube casing.

To streamline the prototyping and testing phase, we combined these six potential solutions together into three designs, with each using one suction cup attachment mechanism and one casing geometry. This resulted in three concepts for further iteration: Twist-fit suction cup attachment with fully-internal casing, press-fit suction cup attachment with box casing, and compliant clip suction cup attachment with original casing, labeled Concept A, B, and C respectively. Based on the requirements in chapter III-A, design priorities were established in the percentage distribution table in Appendix A. Concepts A, B, and C were ranked using a simplified version of this table in Appendix B.

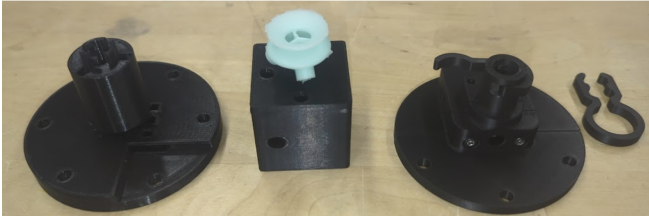


Fig. 3: Early Iterations of Concepts A (Twist-fit/Fully-internal), B (Press-fit/Box), and C (Compliant Clip/Original)

Each concept explored multiple design directions aimed at improving usability and simplicity. We iterated and developed each concept until it achieved basic functionality, at which point we evaluated their performance through hands-on testing (without the robot arm) and against the criteria outlined in Appendix B. Concept C received the highest scores for both the suction cup attachment mechanism and the casing geometry, and was therefore selected as the foundation for further development. The fully-internal design from Concept A was also considered for further development and integration into Concept C.

F. Development Toward Final Design

The finalization of the design can be divided into two parts: The mold used to create the suction cup, and the

casing.



Fig. 4: Improved Mold and Suction Cup

The suction cup mold was re-engineered to increase the success rate of fabrication. With its original design, the suction cups would suffer from frequent tears and punctures during the de-molding process due to the narrow internal chambers. When removing the suction cup, the chamber walls would need to be stretched significantly, leading to high stresses and posing a substantial risk of tearing. To solve this, the inner chamber mold was redesigned to come apart into two parts when de-molding, which facilitated the suction cup's extraction and greatly increased the de-molding success rate.

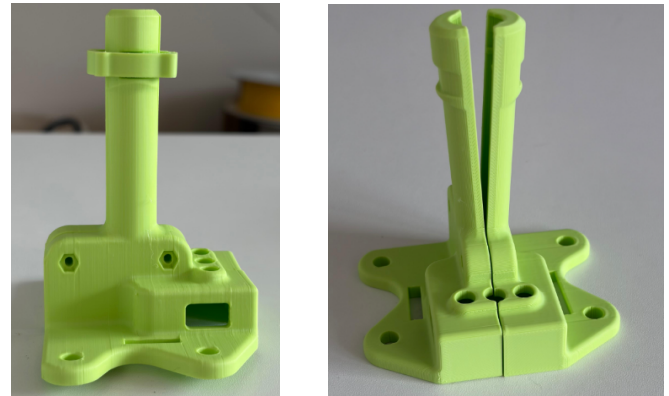
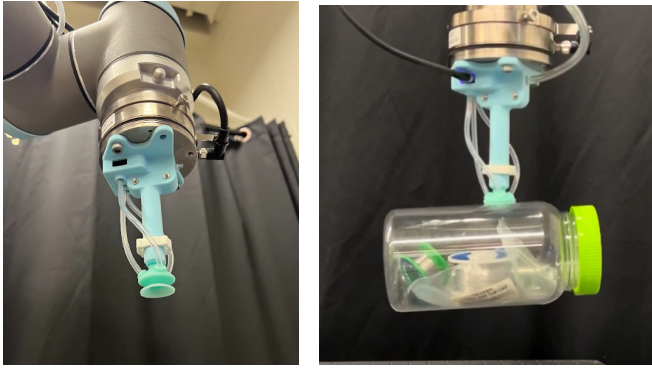


Fig. 5: Final compliant casing design with ring on/off

The casing was further optimized and improved from Concept 3 for greater ease of use and durability, with the most significant change being the integration of a compliant mechanism to attach the suction cup. Instead of having an external compliant clip, this redesign integrated compliance into the casing itself while having an external rigid ring to hold the casing shut. This new compliant casing enabled the rapid removal and insertion of the suction cup without the use of tools, and significantly simplified the system by reducing the overall part count. This mechanism is shown in Fig. 5 above. After testing and evaluating this new design, it was confirmed to have greater performance metrics than any of the previous concepts, and was therefore selected for our final design. Photographs of the fully assembled final prototype mounted on the robot are shown in Fig. 6 below.



(a) Final prototype mounted on robot arm (b) Final prototype grasping transparent, oddly shaped object

Fig. 6: Final multi-chambered suction cup system

IV. PERFORMANCE OF THE FINAL THREE-CHAMBERED SUCTION CUP SYSTEM

The final suction cup design is evaluated using the percentage distribution table from Appendix A as well as the measurable tests defined in the previous chapter.

A. Meeting the Design Requirements in Practice

Table II summarizes the measurable criteria from Section III-A alongside the current test results.

Requirement	Target value	Measured result
Payload	0.5 kg	0.72 kg (Test 1)
Vacuum Performance	80% vacuum at nominal payload	80% vacuum at nominal payload (Test 1)
Holding force margin	$1.2 \times$ payload	$1.44 \times$ payload (Test 1)
Replacement time	≤ 10 min	2 mins 20 secs (Test 2)
Lifetime	≥ 5000 picks	no visible wear before failure (Test 3)

TABLE II: Overview of measurable requirements and current test results

B. Performance Evaluation of the Final Prototype

The final prototype is still in evaluation and performance metrics will be added after statistically significant results have been achieved.

a) *Maximum payload and safety margin:* Using Test 1, the suction cup is characterized over a range of payloads from 10 g to 1.5 kg on representative objects (plastic trays, bottles, metal lids, cardboard pieces). The maximum safe payload is 0.72 kg, corresponding to a safety factor of 1.44 relative to the nominal design payload. No structural failures were observed; grasp failures are dominated by seal loss on highly curved or flexible items.

b) *Setup time and interchangeability:* From Test 2, the complete system setup from a disassembled state requires an average of 10 minutes. Most of the time is spent on firmware configuration, while mechanical assembly is completed 2 minutes 20 seconds, with a standard deviation of 1 minutes across five operators. Component replacement, such as swapping a suction cup membrane or vacuum

tube is much faster, requiring only 30 seconds and can be performed without specialized tools. Variability between operators suggests that assembly documentation and connector labeling are effective in guiding repeatable setup and quick maintenance.

c) *Durability and maintainability:* A full endurance experiment of Test 3 has not been performed yet, the cup body is projected to withstand 5000+ picks in a dusty environment with mixed abrasive and non-abrasive items without noticeable wear in any components before failure, with grasp success rate remaining above 80% after which a gradual decline is observed. Test 1 shows cup replacement, including inspection and filter cleaning, can be performed in 2 minutes 20 seconds, satisfying the maintainability requirement.

C. Weighted-Criteria Evaluation of the Final Prototype

Using a simplified percentage distribution table based on appendix A, the final suction cup system scores 9/10, see Appendix B. A full score using appendix A will be assessed after final testing. The score for this combination of previous concepts scores higher than the separate concepts evaluated in the first design phase and reflects strong improvements in usability and robustness to partial contact, sensing capability, and maintainability, while leaving room for improvement in maximum payload and energy efficiency.

D. Readiness for Deployment in Research and Industry

The final design is easily adaptable for compatibility with different robotic cells and object classes, which is demonstrated by adapting mount geometry to 5 different standard ISO 9409-1 flanges without major redesign of the sensing hardware. Our three-chambered layout provides informative pressure signals that can be used by higher-level control policies for failure detection and haptic exploration, meeting the performance of the research grade Smart Suction Cup Prototype. Overall, the prototype meets or closely approaches the key functional requirements.

V. DISCUSSION: IMPLICATIONS FOR ROBOTIC RECYCLING AND RESEARCH HARDWARE

A. What Tactile Flow Sensing Adds Beyond Vision

The results of the test campaign suggest that chamber-level flow sensing significantly improves robustness compared to purely vision-based suction grasping. Even when only part of the contact area is available, the multi-chamber cup maintains a stable grasp, because individual chambers can still seal and provide informative pressure signals about local contact quality. This enables the system to distinguish between full seals, partial seals, and outright failures, which vision alone cannot reliably infer once the object is occluded by the gripper. In practice, this means that the Smart Suction Cup can recover from visual uncertainty and small pose errors, especially on adversarial objects that are transparent, deformable, or porous.

Beyond robustness, tactile flow sensing also opens the door to richer control strategies. The per-chamber pressure

patterns observed during Tests 2 and 3 encode information about object geometry, stiffness, and leakage paths, which can be exploited for haptic search, online grasp refinement, or automatic rejection of “bad” grasps before a drop occurs. This was already shown in prior work on the research-grade Smart Suction Cup and can now be deployed more broadly. For research labs, having an open, documented implementation of these sensing capabilities lowers the barrier to experimenting with tactile policies on real waste streams and other challenging manipulation tasks.

B. Design Trade-Offs: Compactness, Cost, and Maintainability

The design process revealed several notable trade-offs between compactness, cost, and maintainability. Increasing the number of chambers and sensors improves redundancy and the richness of tactile information but adds complexity to the manifold routing, raises the bill of materials, and can increase the overall volume and weight of the end-effector. Conversely, aggressively minimizing size and part count risks makes the internal tubing difficult to assemble and maintain, or reduces the structural margins needed to safely handle higher payloads. The current prototype represents a compromise: a limited number of chambers (3), a standardized flange interface, and geometry that fits within typical recycling cell envelopes, while remaining manufacturable with 3D-printing and basic laboratory tools.

Maintainability emerged as a particularly important constraint for both labs and industrial users. Design choices such as split housings, accessible PCB mounting, and standardized connectors were necessary to meet the precondition that the tool can be opened, a component replaced, and reassembled within 10 minutes. However, these choices sometimes conflicted with compactness goals or increased the number of printed parts. Similarly, optimizing the connector tube, tubing paths, and cable routing to avoid kinks and accidental disconnections had a direct impact on the external shape of the casing and the ease of printing. As a result, the final design accommodates optional tubing casing if required, while remaining fully functional without it. These trade-offs suggest that future iterations will need to consider not only performance metrics like payload and success rate, but also “time-to-service” and ease of modification as first-class design objectives.

C. Limitations of the Current Prototype

Further testing will drive ongoing iteration and continuous improvement. Despite demonstrating proof of concept, Several limitations of the current prototype can already be acknowledged. First, all tests to date have been conducted in controlled lab conditions that approximate, but do not fully reproduce, the contamination levels, object diversity, and mechanical disturbances present in industrial recycling facilities.

There are also limitations in the mechanical design itself. The present cup body and manifold geometries were tuned for a specific size class of objects and a particular range of

vacuum levels; scaling to heavier items or very small objects may require new cup variants rather than a simple parameter change. Finally, the casing and mounting hardware are modular but still assume a certain style of robot flange and cable management, which may not generalize to all industrial arms without adapters. Addressing these limitations is a key focus of the future work outlined in the subsequent section.

VI. CONCLUSION AND FUTURE DIRECTIONS FOR THE SMART SUCTION CUP PLATFORM

A. Summary of Contributions

The final prototype of the multichamber suction cup satisfies the defined requirements and criteria within the current test campaign, with the exception of the still-unknown lifetime and maximum payload margins, see Table II. The desired robustness under partial contact and surface variability is achieved and will be further improved in ongoing tests, demonstrating clear proof of concept for multi-chambered suction in recycling environments. This indicates that the status quo of simple single-chamber suction grippers is not the only nor the most effective solution for robotic waste sorting.

B. Key Lessons and Design Trade-Offs

The achieved robustness under partial contact and surface variability shows that more informative, fault-tolerant, and maintainable end-effectors are feasible at comparable cost. At the same time, the remaining unknowns in lifetime and maximum payload margins highlight the trade-offs between compactness, material choice, and structural safety factors that must be resolved in future iterations. Our results suggest that careful integration of multi-chambered flow sensing, manufacturability constraints, and modular mounting can yield practical hardware for both research labs and recycling facilities.

C. Future Work for the Smart Suction Cup Platform

Future work should focus on characterizing the lifetime for different maximum payload margins, validating performance in real material recovery facilities, and refining the design to further reduce cost and complexity. By addressing these open points, the Smart Suction Cup System can evolve into a mature, widely deployable platform that makes high-throughput robotic recycling cells more reliable and attractive for facilities with limited resources.

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VII. APPENDIX

Appendix A: Design Priorities & Criteria Deep-dive

Criteria	Weights	Score ranges	Explanation
Sustainability	3%	1-2: >1000 g CO ₂ 3-4: 700–1000 g CO ₂ 5-6: 400–700 g CO ₂ 7-8: 200–400 g CO ₂ 9-10: 0–200 g CO ₂	An estimation of the total carbon footprint of the final product, combining cup body, manifold, casing and tubing. Materials such as polyurethane or silicone elastomers (3–5 kg CO ₂ /kg) and 3D-printed polymers (0.5–1 kg CO ₂ /kg) are considered, together with avoided waste and reusability of components.
Complexity	8%	1-2: >25 parts 3-4: 18–25 parts 5-6: 12–18 parts 7-8: 6–12 parts 9-10: <6 parts	Complexity is defined as the number of distinct parts and mechanisms that must be assembled (cup body, manifold, casing, mount, tubing interfaces, fasteners, PCB fixtures). Lower part count reduces assembly time and potential failure modes.
(Dis)assembly	12%	1-2: >10 min 3-4: 7–10 min 5-6: 5–7 min 7-8: 3–5 min 9-10: <3 min	Time to fully open the tool, access the suction cup, tubing and PCB, and reassemble, without special fixtures. This reflects ease of cleaning, troubleshooting and component replacement during lab work or deployment.
Mounting	6%	1-2: >8 min 3-4: 6–8 min 5-6: 4–6 min 7-8: 2–4 min 9-10: <2 min	Time required to mount or dismount the complete assembly on an ISO 9409-1 robot flange or equivalent interface. Fast, tool-light mounting is preferred to support experiments on multiple robots and layouts.
Production	10%	1-2: 12–15 actions 3-4: 9–12 actions 5-6: 6–9 actions 7-8: 4–6 actions 9-10: 1–4 actions	Production ease is measured by the number of manufacturing actions and processes: FDM printing with or without supports, basic CNC, laser cutting, casting, and assembly steps. Fewer processes and operations reduce lead time and make replication in a typical university makerspace or lab easier.
Cost	9%	1-2: >150 dollars 3-4: 125–149 dollars 5-6: 100–124 dollars 7-8: 60–99 dollars 9-10: <60 dollars	Total cost of materials and purchased components for one complete end-effector (cup, casing, tubing, valves, sensors, electronics). Lower cost supports replication by other labs and deployment in cost-sensitive recycling plants.
Operation Time	5%	1-2: >60 min 3-4: 61–120 min 5-6: 121–180 min 7-8: 181–240 min 9-10: <240 min	Total mass of the cup, mount, casing and integrated hardware. While constrained by robot payload, lower weight improves dynamic performance and makes manual handling safer and more comfortable.
Compactness	8%	1-2: >500 mm 3-4: 400–499 mm 5-6: 300–399 mm 7-8: 200–299 mm 9-10: <200 mm	Measured as overall stack height and bounding box around largest dimensions of the device. A compact design reduces collision risk in cluttered recycling cells and allows closer packing of objects on the conveyor.
Durability	18%	1-2: <5000 picks 3-4: 5000–10000 picks 5-6: 10000–20000 picks 7-8: 20000–40000 picks 9-10: >40000 picks	Expected number of successful picks before the cup body or mechanical mount must be replaced. Durability combines wear of the sealing lip, fatigue of compliant features, and robustness of tubing interfaces under typical recycling duty cycles.
Grasp robustness	21%	1-2: <30% success 3-4: 30–50% success 5-6: 50–65% success 7-8: 65–80% success 9-10: >80% success	Most critical performance criterion: success rate in different situations. It captures tolerance to leaks, geometry variation and contamination, and is directly linked to robotic performance in Material Recovery Facilities.
Total	100%		

TABLE III: Importance of criteria and rating scales for the suction cup design

Appendix B: Rough concepts rating with criteria

Criteria	Weight (%)	Twist-Fit	Press-fit	Clip	Final
Lifetime	20	6	4	4	6
(Dis)assembly and cleaning access	40	8	8	10	10
Simplicity / Usability	25	2 (not airtight)	4	4	10
Production (e.g., 3D print supports)	15	8	8	10	10
Weighted total (100%)		5.66	6.2	7.3	9

TABLE IV: Weighted decision matrix comparing three design concepts, with the final chosen concept shown on the right.

Notes: We concluded the Twist-Fit will not meet requirements for functionality. The Press-fit concept gives limited access to tubing and scores low on usability. The Clip concept will be used as a building block and desirable features from multiple concepts will be integrated into the final concept starting from there.

Appendix C: final design Bill Of Materials

TABLE V: Combined bill of materials for the final suction cup system

Part	Qty	Specification	Unit (USD)	Price	Total Price (USD)
Casing Left	1	PLA, 18.446 cm ³ , 22.873 g	N/A		0.80
Casing Right	1	PLA, 18.290 cm ³ , 22.680 g	N/A		0.79
Ring	1	PLA, 1.273 cm ³ , 1.579 g	N/A		0.055
Vacuum tube	1	Main vacuum line	0.05		0.05
Pressure tube	3	Polyurethane sensing tubes	0.5		0.15
Plastic adapter/connector	2	Tube fittings / hose adapters	0.03		0.06
M3 screw	2	Assembly fasteners	0.03		0.06
Metal rods mold	4	Suction cup mold hardware	0.1		0.4
Mold M3 screws	2	Fasteners for suction cup mold	0.03		0.06
C1	1	1 μ F capacitor, C.0201.0603Metric	0.08		0.08
C2, C3, C4	3	1000 pF capacitor, C.0402.1005Metric	0.08		0.24
R1–R8	8	4.7 k Ω resistor, R.0201.0603Metric	0.10		0.80
R9	1	10 k Ω resistor, R.0603.1608Metric	0.10		0.10
U1	1	XIAO-RP2040-SMD	4.68		4.68
U2	1	TCA9548ARGER	1.10		1.10
U3, U4, U5	3	MPRLS0025PA00001 pressure sensor	4.0300		12.08
PCB subtotal					18.98
PCB assembly subtotal (scales down with volume)					23.7
PCB Shipping and additional costs					scales down with volume
Mechanical subtotal					2.965
Total					45.645